Association of Food Consumption and Child Stunting

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Overall structure of my thesys

1. Background
2. Research interest
3. Factor analysis of stunting reduction focusing on food consumption (IV: CCT program participation rate)
4. Impact of multisector intervention on stunting reduction
5. 2SLS focusing on interaction
6. Impact of CRECER program in stunting reduction
7. PSM-DID for impact assessment of CRESER program
8. Mediation analysis to assess contribution of CRECER program
9. Discussion

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Section 1

Background

Current status

- Malnutrition is one of the world's most serious but least-addressed development challenges. Its human and economic costs are enormous, falling hardest on the poor, women, and children.
- South Asia and Sub-Saharan Africa are the hardest hit, with with roughly two-thirds of the 150.2 million stunted children residing in these two regions(World bank).
- The economic costs of undernutrition are significant with at least $1 trillion a year because of productivity loss due to undernutrition and micronutrient deficiencies (FAO 2013).
- From meta-analysis of 37 international study, it is revealed that stunting was strongly linked to household food insecurity and poor dietary diversity (2025, islam et. al).
- Meanwhile, it is well known that other various factors including mother's education status, access to clean water and toilet, economic disparity, fragile agro-ecosystem have also significant impact on malnutrition.

Why we need multisector intervention?

three key factors are associated with undernutrition

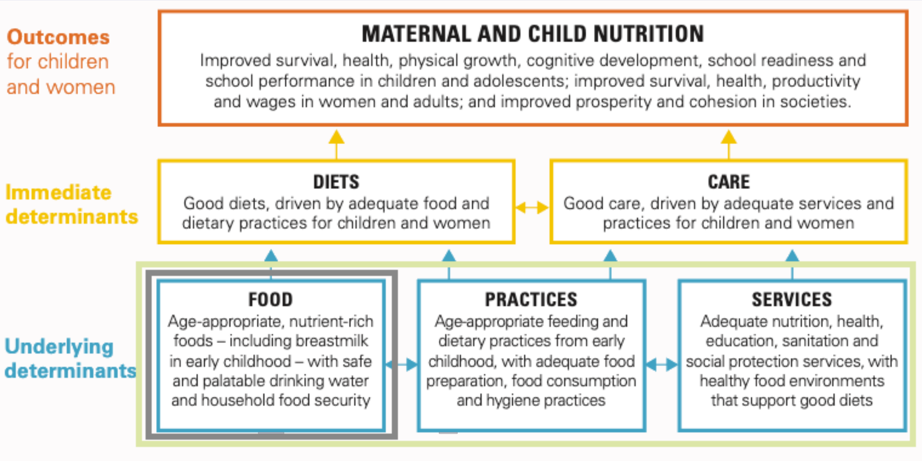

Figure 1: UNICEF framework of undernutrition

Peru case study

Key factors associated with undernutrition in Peru

Supplement

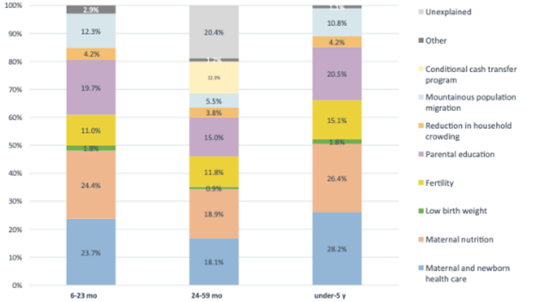

FIGURE 6 Decomposing predicted changes in HAZ (i.e., percentage contribution of determinant domains) for children under-5 from 2000 to 2016. The <6 mo age category results are not presented owing to small sample sizes and minimal HAZ changes in this population causing unstable parameters. Indicators included maternal and newborn health care (skilled birth attendant and antenatal care ≥4), low birth weight, parental education (maternal and paternal education), mountainous population migration (altitude), maternal nutrition (maternal BMI and height), fertility (parity and interpregnancy interval), reduction in household crowding (number of household members), and conditional cash transfer program (Juntos). "Other" category includes child age, sex, and region for 6- to 23-mo and 24- to 59-mo age groups in addition to economic improvement (wealth index) (0.3%) and childhood vaccines (measles) (0.4%) for the 6- to 23-mo age group, and maternal age (older age pregnancy) (0.5%) and reduction in diarrhea incidence (0.6%) for children under-5. Parental education breakdown: for children 6–23 mo: maternal: 17.5%, paternal: 2.2%; children 24–59 mo: maternal: 13.1%, paternal: 1.9%; and children under-5: maternal: 17.8%, paternal: 2.7%. HAZ, height-for-age *z* score.

Figure 2: Drivers of stunting reduction in Peru: a country case study (Luis, et. Al, 2020)

Different type of dietary guidance

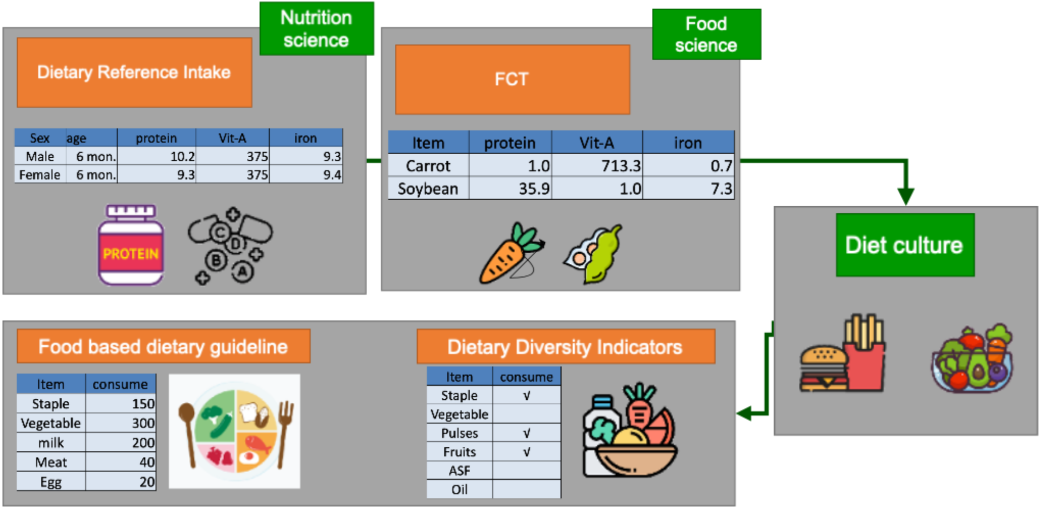

Figure 3: different type of dietary guideline

How to evaluate quality of diet?

Table 1: Key consideration in evaluating quality of diet

Key consideration in evaluating quality of diet	
Diversity	Diversity of **food group** (qualitative)
Balance	Consumed **food group** balance (semi-quantitative)
Adequacy	Adequacy of key **nutrient intake** from diet (quantitative)
Moderation	Refrain **overconsumption**

Is Dietary Diversity Score effective tool to assess diet quality?

- Ruel conducted intensive review of 47 articles to examine impact of various agriculture intervention to nutrition improvement. Overall, number of studies showed clear impact of NSA to DDIs, while impact on nutrition outcome was limited (Ruel, 2018)
- Eric conducted peer review from 137 articles to examine relationship between DDIs and health impact, found that associations between DDIs and health outcomes were largely inconsistent (Eric, 2021)

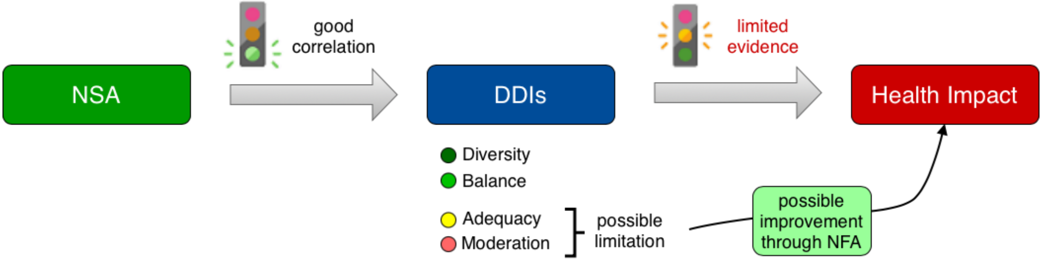

Figure 4: effectiveness of DDIs

Descriptive statistics1

Table 2: descriptive statistics-1

Characteristic	N	**control_pre(<2009)** N = 21,836[1]	**treat_pre** N = 8,181[1]
Height under 145cm - women	30,017	2,426 (11%)	1,264 (15%) *
Birth weight less than 2.5 kg	30,017	1,289 (5.9%)	454 (5.6%)
more than 3 child under 12	30,017	3,140 (14%)	1,708 (21%) *
secondary or higher education	30,017	11,414 (52%)	2,334 (29%) ***
own a sheep	30,017	2,950 (14%)	1,900 (23%) **
Skilled assistance during ANC	30,017	9,196 (42%)	4,391 (54%) ***
owns refrigerator	30,017	7,023 (32%)	818 (10.0%) ***
Overweight or obese BMI - women	30,017	11,309 (52%)	3,413 (42%) **

[1] n (%); Mean (SD)

Descriptive statistics2

Table 3: descriptive statistics-2

Characteristic	N	control_pre(<2009) N = 21,836[1]	treat_pre N = 8,181[1]
water fetching over 30min	30,017	74 (0.3%)	134 (1.6%) *
HH head over 29	30,017	17,560 (80%)	6,181 (76%)
living in capital city	30,017	3,325 (15%)	0 (0%) ***
open defication	30,017	3,301 (15%)	1,487 (18%)
using traditional cooking fuel	30,017	8,437 (39%)	6,204 (76%) ***
wealth index (normalized score)	30,017	0.44 (0.82)	-0.44 (0.76) ***
Problem health care access: permission to go	30,017	3,054 (14%)	1,195 (15%)
First marriage by age 15	30,017	1,439 (6.6%)	576 (7.0%)

[1] n (%); Mean (SD)

Descriptive statistics3

Table 4: descriptive statistics-3

Characteristic	**N**	**control_pre(<2009)** N = 21,836[1]	**treat_pre** N = 8,181[1]
Treated water by straining through cloth before drinking	30,017	12 (<0.1%)	0 (0%)
stool improper handling	30,017	14,560 (67%)	5,498 (67%)
Agree that husband is justified in hitting or beating his wife for at least one of the reasons	30,017	1,056 (4.8%)	661 (8.1%) *
Child given grains in day/night before survey- last-born under 2 years	30,017	20,488 (94%)	7,100 (87%) ***
Child given roots or tubers in day/night before survey- last-born under 2 years	30,017	17,688 (81%)	7,257 (89%) ***
Child given vitamin A rich food in day/night before survey- last-born under 2 years	30,017	16,453 (75%)	5,402 (66%) **

[1] n (%); Mean (SD)

Descriptive statistics4

Table 5: descriptive statistics-4

Characteristic	N	**control_pre(<2009)** N = 21,836[1]	**treat_pre** N = 8,181[1]
Child given other fruits or vegetables in day/night before survey- last-born under 2 years	30,017	14,909 (68%)	5,319 (65%)
Child given legumes or nuts in day/night before survey- last-born under 2 years	30,017	9,694 (44%)	4,344 (53%) *
Child given meat, fish, shellfish, or poultry in day/night before survey- last-born under 2 years	30,017	18,314 (84%)	5,048 (62%) ***
Child given eggs in day/night before survey- last-born under 2 years	30,017	10,986 (50%)	5,507 (67%) ***
Child given cheese, yogurt, or other milk products in day/night before survey- last-born under 2 years	30,017	6,972 (32%)	2,574 (31%)
Child with minimum dietary diversity, 5 out of 8 food groups- last-born 6-23 months	30,017	18,043 (83%)	6,226 (76%) **

[1] n (%); Mean (SD)

Descriptive statistics5

Table 6: descriptive statistics-5

Characteristic	N	**control_pre(<2009)** N = 21,836[1]	**treat_pre** N = 8,181[1]
Child with minimum meal frequency- last-born 6-23 months	30,017	19,484 (89%)	7,616 (93%) *
Stunted child under 5 years	30,017	4,865 (22%)	2,877 (35%) ***
Wasted child under 5 years	30,017	234 (1.1%)	38 (0.5%)
Any anemia - child 6-59 months	30,017	14,997 (69%)	5,228 (64%)

[1] n (%); Mean (SD)

Descriptive statistics6

Table 7: descriptive statistics-1

Characteristic	N	control_post(>=2009) N = 117,260[1]	treat_post N = 17,467[1]
Height under 145cm - women	134,726	11,046 (9.4%)	2,301 (13%) ***
Birth weight less than 2.5 kg	134,726	6,734 (5.7%)	1,410 (8.1%) ***
more than 3 child under 12	134,726	10,551 (9.0%)	2,382 (14%) ***
secondary or higher education	134,726	72,629 (62%)	6,347 (36%) ***
own a sheep	134,726	8,770 (7.5%)	3,320 (19%) ***
Skilled assistance during ANC	134,726	50,627 (43%)	7,633 (44%)
owns refrigerator	134,726	59,703 (51%)	3,192 (18%) ***
Overweight or obese BMI - women	134,726	66,871 (57%)	8,021 (46%) ***

[1] n (%); Mean (SD)

Descriptive statistics2

Table 8: descriptive statistics-2

Characteristic	N	control_post(>=2009) N = 117,260[1]	treat_post N = 17,467[1]
water fetching over 30min	134,726	2,001 (1.7%)	143 (0.8%) **
HH head over 29	134,726	92,902 (79%)	12,971 (74%) ***
living in capital city	134,726	39,064 (33%)	0 (0%) ***
open defication	134,726	11,045 (9.4%)	2,511 (14%) ***
using traditional cooking fuel	134,726	24,990 (21%)	10,715 (61%) ***
wealth index (normalized score)	134,726	0.72 (3.44)	-1.30 (3.60) ***
Problem health care access: permission to go	134,726	14,255 (12%)	2,638 (15%) ***
First marriage by age 15	134,726	4,216 (3.6%)	925 (5.3%) ***

[1] n (%); Mean (SD)

Descriptive statistics3

Table 9: descriptive statistics-3

Characteristic	N	control_post(>=2009) N = 117,260[1]	treat_post N = 17,467[1]
Treated water by straining through cloth before drinking	134,726	84 (<0.1%)	0 (0%) *
stool improper handling	134,726	97,858 (83%)	12,261 (70%) ***
Agree that husband is justified in hitting or beating his wife for at least one of the reasons	134,726	3,932 (3.4%)	697 (4.0%)
Child given grains in day/night before survey- last-born under 2 years	134,726	105,855 (90%)	15,459 (89%) *
Child given roots or tubers in day/night before survey- last-born under 2 years	134,726	93,083 (79%)	14,613 (84%) ***
Child given vitamin A rich food in day/night before survey- last-born under 2 years	134,726	92,801 (79%)	12,741 (73%) ***

[1] n (%); Mean (SD)

Descriptive statistics4

Table 10: descriptive statistics-4

Characteristic	**N**	**control_post(>=2009)** N = 117,260[1]	**treat_post** N = 17,467[1]
Child given other fruits or vegetables in day/night before survey- last-born under 2 years	134,726	84,968 (72%)	12,120 (69%) **
Child given legumes or nuts in day/night before survey- last-born under 2 years	134,726	48,559 (41%)	8,870 (51%) ***
Child given meat, fish, shellfish, or poultry in day/night before survey- last-born under 2 years	134,726	100,408 (86%)	12,385 (71%) ***
Child given eggs in day/night before survey- last-born under 2 years	134,726	66,384 (57%)	10,865 (62%) ***
Child given cheese, yogurt, or other milk products in day/night before survey- last-born under 2 years	134,726	49,842 (43%)	7,125 (41%)
Child with minimum dietary diversity, 5 out of 8 food groups- last-born 6-23 months	134,726	92,771 (79%)	13,408 (77%) *

[1] n (%); Mean (SD)

Descriptive statistics5

Table 11: descriptive statistics-5

Characteristic	N	control_post(>=2009) N = 117,260[1]	treat_post N = 17,467[1]
Child with minimum meal frequency- last-born 6-23 months	134,726	108,942 (93%)	16,456 (94%) *
Stunted child under 5 years	134,726	16,081 (14%)	5,290 (30%) ***
Wasted child under 5 years	134,726	795 (0.7%)	113 (0.6%)
Any anemia - child 6-59 months	134,726	64,743 (55%)	10,302 (59%) **

[1] n (%); Mean (SD)